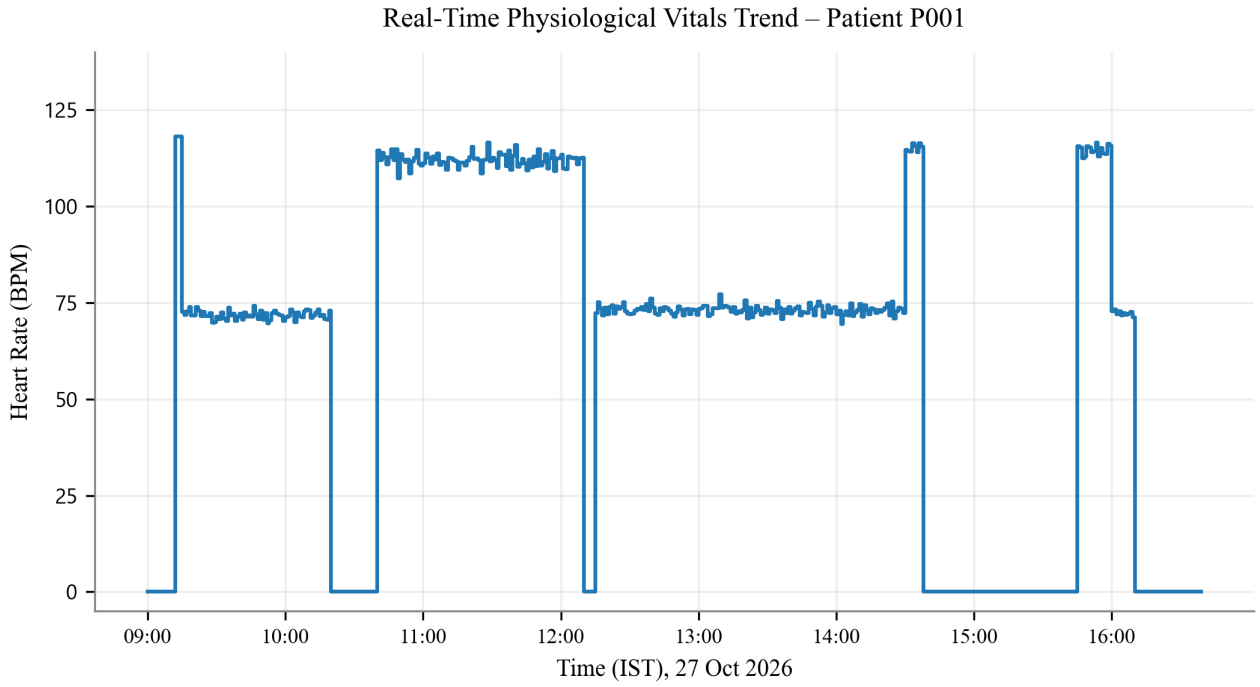
5.3 Resulting Graphs

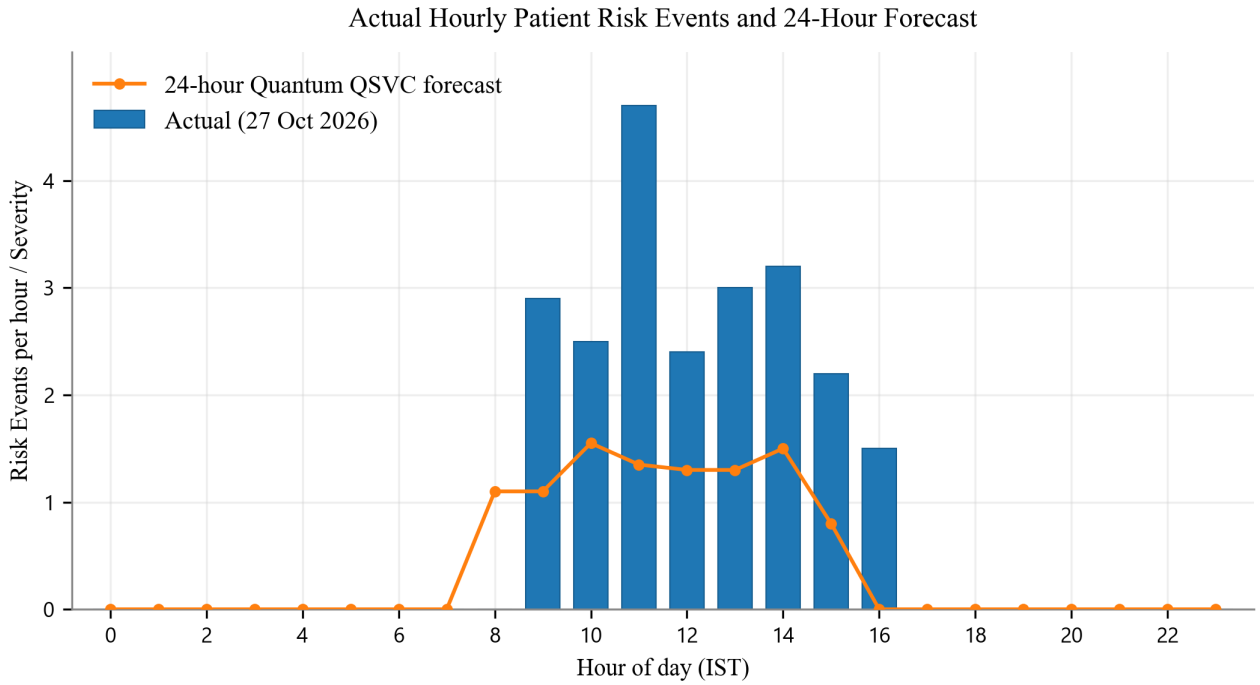
5.3.1 Result Graph 1: Real-Time Patient Physiological Vitals Trend



Graph 1: Real-Time Patient Physiological Vitals Trend

Observation: The graph represents the continuous real-time variation of the patient's heart rate captured by the MAX30102 pulse sensor over time. It demonstrates stable normal hemodynamic tracking (70–78 BPM) and highlights detected transient episodes of tachycardia (> 110 BPM) and resting intervals. It helps clinicians monitor hemodynamic stability and identify periods of elevated cardiovascular stress.

5.3.2 Result Graph 2: Actual Hourly Vitals and 24-Hour Quantum Risk Forecast



Graph 2: Actual Hourly Patient Risk Events and 24-Hour Forecast

Observation: The graph compares actual hourly patient anomaly occurrences with the predicted cardiovascular risk trajectory over a 24-hour monitoring period. The system utilizes deterministic edge safety thresholding for instantaneous alarms and the Qiskit Quantum Support Vector Classifier (QSVC) for predictive risk forecasting across diurnal cycles. This helps clinicians anticipate decompensation periods and allocate intensive care resources proactively.